Akshay Shankar

COMPUTATIONAL PHYSICIST

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Research Focus: *Understanding the behaviour of ultra-cold matter through numerical simulations to enable practical quantum technologies.*

Summary

PhD Student at Ghent University specializing in Tensor Network algorithms for one-dimensional quantum gases in the continuum. Experienced in developing open-source software for high-performance numerical algorithms (Julia/Python) to study quantum many-body systems. Also experienced in building interactive visualizations and user interfaces using front-end web development tools and game engines.

Skills

Scientific Programming Julia (TensorKit.jl, MPSKit.jl), Python (NumPy, SciPy), C++, Git, Linux

Numerical Methods Tensor Networks (DMRG, VUMPS, TDVP), Quantum Monte Carlo (SSE), Mean-field theory (Gutzwiller, Cluster-MFT)

Visualization & UI Godot Engine (GDScript), Blender, Inkscape, Front-end (HTML, CSS, JavaScript)

Research Experience

Ph.D. Project: Reconstructing quantum dynamics in a synthetic dimension of trap states *Ghent University*

SUPERVISED BY PROF. KAREL VAN ACOLEYEN | PYTHON *Oct. 2023 - Present*

- Collaborating with the Cold Atoms group in Gent to characterize the quantum dynamics in a Floquet-engineered synthetic lattice realized in a Rubidium-87 experimental setup.
- Developed a Python-based data-processing pipeline to perform quantum state tomography, converting raw absorption imaging data into reconstructed density matrices in the synthetic dimension basis.

Ph.D. Project: Finite-element matrix product states for continuum models *Ghent University*

SUPERVISED BY PROF. KAREL VAN ACOLEYEN AND PROF. JUTHO HAEGEMAN | JULIA *Oct. 2023 - Present*

- Developed a theoretical framework to utilize Matrix Product States (MPS) with *non-orthogonal* single-particle orbitals to study quantum gases in one dimension.
- Applied the framework to capture ground state physics of the Lieb-Liniger model using a finite-element basis and a multi-grid optimization scheme to efficiently reach the continuum limit.
- Developed a performant, open-source numerical implementation of the finite-element Matrix Product States (FE-MPS) formalism involving custom DMRG code built on top of MPSKit.jl.

M.S. Thesis: Low-temperature phases of interacting bosons in a lattice *IISER Mohali & University of Stuttgart*

SUPERVISED BY DR. SANJEEV KUMAR AND PROF. TILMAN PFAU | JULIA, PYTHON *Jun. 2022 - Apr. 2023*

- Implemented various many-body algorithms (Cluster Mean-Field Theory, Stochastic Series Expansion, Neural Quantum States) to map the phase diagram of the extended Bose-Hubbard model on a square lattice.
- Collaborated with experimentalists at the University of Stuttgart to guide the exploration of supersolid phases arising from Dysprosium atoms with dipole-dipole interactions in an optical lattice.

Research Internship: Simulating BEC dynamics in one dimension *IIT Kharagpur (Remote)*

SUPERVISED BY DR. VISHWANATH SHUKLA | JULIA, FORTRAN90 *May 2021 - Aug. 2021*

- Developed a 1D Gross-Pitaevskii Equation (GPE) solver to simulate ground-state physics and real-time dynamics of Bose-Einstein Condensates.
- Investigated parallel computing frameworks (MPI, OpenMP) to optimize performance.

Publications & Preprints

Finite-Element Matrix Product States for Continuum Models in One Dimension *Arxiv*

AKSHAY SHANKAR, KAREL VAN ACOLEYEN, AND JUTHO HAEGEMAN *Jun. 2026*

https://arxiv.org/abs/2606.14873

Collaborative projects

Game Jam Competitions

TEAM LEAD & LEAD PROGRAMMER | GODOT, GIT *2023 - Present*

- Led teams of up to 5 to conceptualize and develop interactive game prototypes within strict 48–96 hour deadlines.
- Utilized Git for version control, ensuring project stability under rapid development cycles.
- Managed the programming life-cycle from designing the overall code architecture to final deployment on itch.io through Github Actions.

ThePhysicsHub

CO-FOUNDER | JAVASCRIPT, P5.JS *2020-2021*

- Founded a global community of students to maintain a repository of interactive simulations for educational purposes.
- Established a standardized user interface and developed several simulations demonstrating concepts of classical mechanics.

Open Source Software

TentMPS.jl

IMPLEMENTATION OF THE FE-MPS FORMALISM DEVELOPED IN ARXIV:2606.14783. *2026-Present*

LiebLinigerBetheAnsatz.jl

IMPLEMENTATION OF THE BETHE ANSATZ SOLUTION FOR THE LIEB-LINIGER MODEL. *2025-Present*

Education

Ghent University *Gent, Belgium*

PH.D. IN PHYSICS *Oct. 2023 - Present*

- Supervised by Prof. Karel Van Acoleyen and Prof. Jutho Haegeman.

Indian Institute of Science Education and Research (IISER), Mohali *Punjab, India*

B.S.-M.S. INTEGRATED DEGREE (MAJOR IN PHYSICS) *Aug. 2018 - June. 2023*

- **Cumulative Performance Index (CPI): 10.0/10.0**

Honors & Awards

2023	**President's Gold Medal**, for the best academic performance in the class of 2023.	*IISER Mohali*
2019	**CNR Rao Foundation Award**, for obtaining 10.0 SPI in the 2nd semester.	*IISER Mohali*
2018	**Innovation in Science Pursuit for Inspired Research - Scholarship for Higher Education (INSPIRE-SHE)**, for top 1% performance in AISSCE.	*DST India*

Conferences & Workshops

2026	**Challenges of Low-Dimensional Quantum Gases: Brazilian-German Wilhelm and Else Heraeus Seminar**, presented a poster titled "Finite element matrix product states for continuous systems"	*ICTP-SAIFR, Sao Paulo, Brazil*
2024	**Les Houches Predoc school on cold atoms**, presented a poster titled "RadialMPS: A novel scheme to study 2D Bose gases in the continuum"	*Les Houches, France*
2024	**European tensor network school**, presented a poster titled "RadialMPS: A novel scheme to study 2D Bose gases in the continuum"	*Padua, Italy*